Why $m_{ij}(t) = 1$ does not give $m_{ji}(t) = 1$

The two halves of the pair contact, with a worked example

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Companion of the note *Group flights in free flight* (same folder)

The answer. No. $m_{ij}(t)$ asks whether i, at instant t, is at a place that j visits within δ; $m_{ji}(t)$ asks whether j, at the same instant, is at a place that i visits within δ. These are two questions about two different pairs of instants. Both are read from one set of matched pairs $\mathcal{M}_{ij}$, one on each axis, and they agree only if that set is symmetric about the diagonal $t' = t$.

What does hold.

- If $m_{ij}(t) = 1$ through the pair (t, t'), then $m_{ji}(t') = 1$: the same pair, read on the other axis, $l = t' - t$ later.
- Every instant of one indicator lies within δ of an instant of the other.
- The contact $a_{ij} = m_{ij} \vee m_{ji}$ is symmetric: $a_{ij} = a_{ji}$.

1 The definitions

For two flights i and j, $\mathcal{M}_{ij}$ is the set of correspondences of the companion note: the pairs of instants (t, t') with $|t' - t| \le \delta$ at which i at t and j at t' are closer than ε_h. The vertical and heading conditions only remove pairs and are left out here. The note defines

$$m_{ij}(t) = \mathbb{1}\big[\exists\, t' :\ (t, t') \in \mathcal{M}_{ij}\big], \qquad m_{ji}(s) = \mathbb{1}\big[\exists\, t :\ (t, s) \in \mathcal{M}_{ij}\big], \qquad a_{ij}(t) = m_{ij}(t) \vee m_{ji}(t).$$

The first fixes the instant of i and lets that of j vary within $\pm\delta$; the second fixes the instant of j and lets that of i vary. Nothing makes the two answers equal.

2 A worked example

Take $\delta = 120$ s and $\varepsilon_h = 500$ m, and ignore heights and headings. Pilot i circles in a thermal at the point P from 0 to 600 s; the radius of the circle is negligible against ε_h. Pilot j flies east in a straight line at 10 m/s and is in P at 500 s.

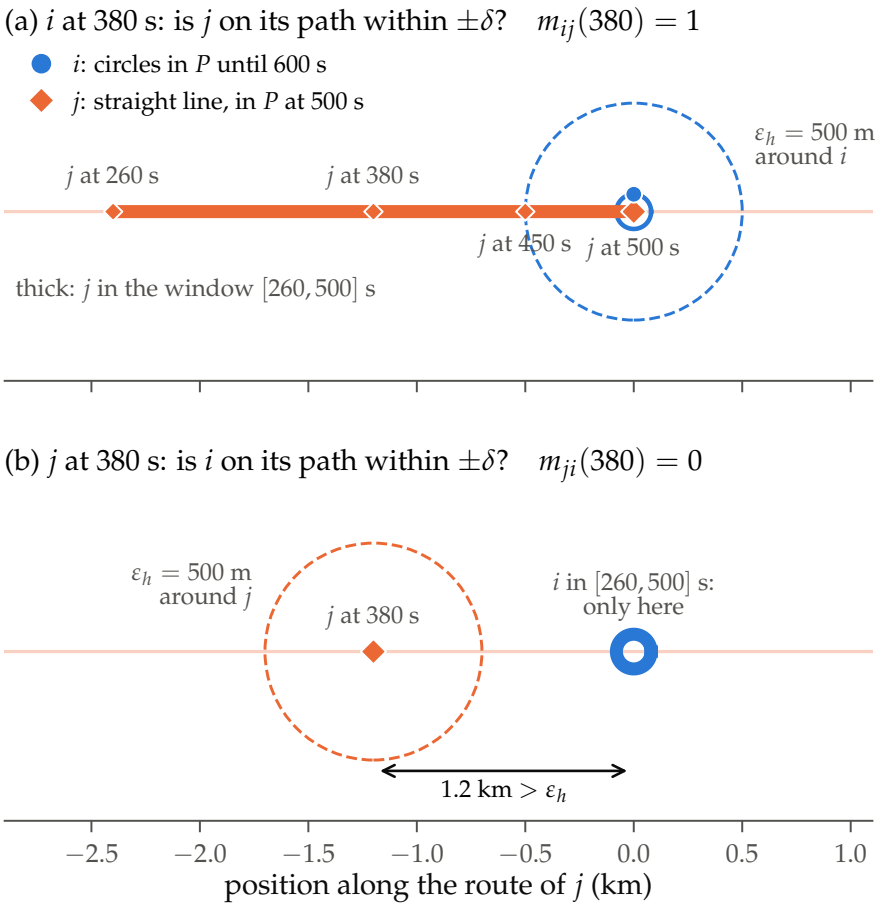


Figure 1: The example at 380 s. (a) i is in P and j is 1.2 km west of it. In the window $[260, 500]$ s of $\pm\delta$ (thick), j comes within ε_h of i at 451 s and is in P at 500 s: the pair $(380, 500)$ is a correspondence, so $m_{ij}(380) = 1$. (b) At the same instant j is 1.2 km from P, and during its window i has been only in P: no position of i lies within ε_h of j, so $m_{ji}(380) = 0$. The ring only marks P.

t (s)	j relative to P	m_{ij}	m_{ji}	a_{ij}	what happens
300	2.0 km west	0	0	0	j needs more than 120 s to come within 500 m of P
380	1.2 km west	1	0	1	i is where j will be 120 s later; j is not yet where i has been
500	in P	1	1	1	both are in P
560	0.6 km east	1	0	1	i is where j was 60 s earlier; j has left

Table 1: The four instants of the example. Over the whole flight $m_{ij} = 1$ from 331 to 600 s, $m_{ji} = 1$ from 451 to 549 s, and $a_{ij} = 1$ from 331 to 600 s (strict inequality: exactly 500 m does not count).

3 The same example in the plane (t, t')

Plot every correspondence (t, t') with the instant at which i passes across and the one at which j passes up (Figure 2a). Then $m_{ij}(t) = 1$ exactly when the *vertical* line at t meets $\mathcal{M}_{ij}$, and $m_{ji}(t) = 1$ exactly when the *horizontal* line at height t meets it. The two indicators are the shadows of $\mathcal{M}_{ij}$ on the two axes.

At $t = 380$ the vertical line meets $\mathcal{M}_{ij}$ (at the pair $(380, 500)$ and the ones just below it), and the horizontal line does not: $\mathcal{M}_{ij}$ has no point with $t' < 451$. The set is not symmetric about the diagonal. The mirror of the pair $(380, 500)$ would be $(500, 380)$, which needs i at 500 s within 500 m of j at 380 s, and that is 1.2 km away.

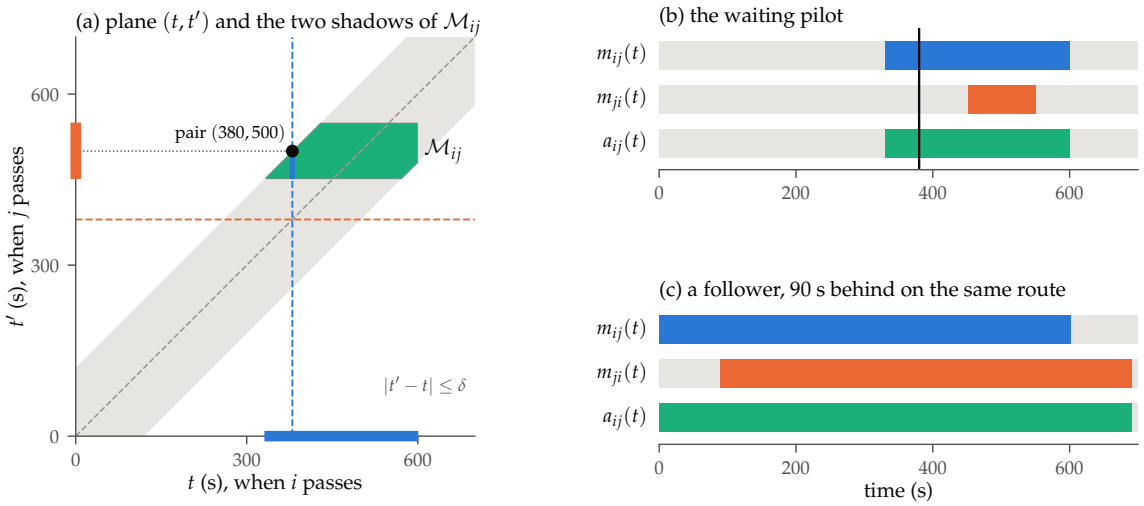


Figure 2: (a) The correspondences $\mathcal{M}_{ij}$ (green) in the band $|t' - t| \le \delta$, and its shadows m_{ij} (blue bar on the t axis) and m_{ji} (orange bar on the t' axis). The vertical line at 380 s meets $\mathcal{M}_{ij}$ (thick segment), the horizontal one does not. (b) The indicators over time, with the line at 380 s. (c) A pilot j on the same route 90 s behind i: m_{ji} is m_{ij} delayed by exactly 90 s.

4 How far apart they can be

Claim. If $m_{ij}(t) = 1$ there is an instant t' with $|t' - t| \le \delta$ and $m_{ji}(t') = 1$; and conversely.

Proof. Take the pair $(t, t') \in \mathcal{M}_{ij}$ that gives $m_{ij}(t) = 1$. Then $|t' - t| \le \delta$ by the definition of $\mathcal{M}_{ij}$, and $m_{ji}(t') = 1$ by the definition of m_{ji}. The converse is the same argument read the other way. □

So the two indicators can differ only by a shift of at most δ. The bound is reached in the example: the first instant of m_{ij} (331 s) is exactly δ before the first instant of m_{ji} (451 s). The union a_{ij} inherits it: at each end of a contact it extends the shorter indicator by up to δ.

5 Why the contact is the union

definition	symmetric	in the example	remark
$m_{ij} \vee m_{ji}$ (used)	yes	331–600 s	counts the approach and the departure within δ
$m_{ij} \wedge m_{ji}$	yes	451–549 s	shorter: each on the path of the other at once
m_{ij} alone	no	331–600 s, or 451–549 s for (j, i)	depends on which pilot is called i

Table 2: Three ways to define the contact of a pair.

The union makes the contact independent of the order of the two pilots and keeps the approach and the departure, where one of them is already on the path of the other. The price is that a contact starts up to δ before the pilots are close at the same time. The mutual version $m_{ij} \wedge m_{ji}$ would give shorter contacts: the false discovery rate and τ^* would have to be computed again with the null model, which is not done here.

In the code. `contacts.scan` returns the instants of $m_{ij} = 1$ and, separately, those of $m_{ji} = 1$; `contacts.pair_intervals` takes their union. Scenarios: `models.WAIT`, `models.FOLLOW`; figures: `plots.asymmetry`; numbers checked in `test_note.py`.